AVEX ESP models

1) Activations (easy - low hanging fruit)
   Produce activation statistics and compare different models and dataset.
   Models: the 4 transformer models
   Dataset: EarthSpeciesProject/NatureLM-audio-training, split by source_dataset

   Step 1: collect all residual stream activations (embeddings, post-layer 0 .. 11) for a significant chunk of NatureLM-audio-training; each dataset item will be a 13 x 513 x 768 torch array. Keep meta-data labels, in particular file_name/metadata (flatten if possible) / Source_dataset / id / output / task
   Q: where do we store activations?

   Step 2: measure activation statistics:
   Activation L2 norms
   Singular values
   PCA alignment
   Intrinsic dimensionality

   Compare across layers and models
   Either use all activations, or mean-pool activations to get one vector per item
   Compare nature sounds to other sound

   Step 3: specific cases
   - Look at dynamics of activations upon addition of white noise / pink noise at different levels; mean-pool activation of a single dataset and repeat for several datasets; find "noise" subspace (Sid)
   - Add two audio sources together; dynamics of representations, how do they "bridge" between one and the other

   Other ideas: barycenter of each species…
   do we see hierarchical representations? See Victor Veich paper

2) Probes (medium) + attribution (harder)
   Train probes at each layer to separate, say, two species of birds; then apply attribution techniques to to recover which patterns they are sensitive to

   Step 1: identify in EarthSpeciesProject/NatureLM-audio-training subsets that correspond to 2 (or more) distinct species. They need to have a substantial amount of items for each species.

   Step 2: train probes (linear probes to start with) to separate one vs the other. Evaluate accuracy across layers.

Step 3: use attribution methods to recover which input patches they rely one. More details coming soon.
Step 4: probe for different separation, for example:
Class: Aves / Mammalia
Order: Passeriformes / Charadriiformes / Piciformes / Strigiformes
Species: a lot
Can we find directions corresponding to say, animal vs music, Aves vs Mammalia, etc.
Do we see hierarchical geometry corresponding to representations?
Apply the gaussian blob fit?
Note: can do that with activation centroids too! (probably easier)

3) Dictionary learning / SAEs (hard)
Might do later, but not priority.